

Leveraging Multisource Label Learning for Underground Object Recognition

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Abstract—Currently, numerous deep learning (DL) methods have been proposed for the recognition of ground penetrating radar (GPR) B-scan images. Due to the sensitivity of GPR imaging to local underground conditions, DL models trained in other underground environment areas are likely to fail in new areas. Consequently, organizing and labeling new GPR images to train models have become a widely adopted approach in practical applications. However, expert annotation is costly, making it often difficult to collect large-scale datasets with high-quality annotations. Some studies have attempted to improve the quality of image annotations by integrating the efforts of multiple annotators. This process can be constrained by the varying levels of expertise and attention spans of the annotators, which may lead to the presence of errors or contradictions in the provided labels. To address these challenges, this article proposes a method for underground target recognition aimed at multisource annotation tasks. A probability multisource label aggregation (PMLA) module is designed to estimate the reliability of multisource labels, and a label-sensitive regularization (LSR) module is introduced to mitigate the negative impact of potentially erroneous labels on model training. Extensive experiments are conducted on multiple GPR B-scan datasets. The experimental results demonstrate the advantages of the proposed method in handling annotation conflicts and improving the accuracy of underground target recognition.

Index Terms—Buried object detection, data processing, GPR data annotation, ground penetrating radar (GPR).

NOMENCLATURE

$\mathbf{X}^k$	Input of GPR B-scan images with known ground truth.
$\mathbf{X}^u$	Input of GPR B-scan images without ground truth.
$\mathcal{C}$	Preset object classes of GPR images.
gt_j^k	Ground truth of x_j^k .
x_j^u	j th image in $\mathbf{X}^u$.
a_i	i th annotator.
p_i	Professional level of annotator a_i .

$k_{i,j}$	Annotation results of x_j^k by annotator a_i .
$u_{i,j}$	Annotation results of x_j^u by annotator a_i .
$\mu_{i,j}^l$	Reliability of label $u_{i,j}$ in the l th iteration.
$\sigma(\mathbf{q}_j)$	Vector of probabilities that image j belongs to each class.
t	Amount of images in $\mathbf{X}^k$.
m	Amount of images in $\mathbf{X}^u$.
r	Amount of object classes.
n	Amount of annotators.
$\mathbf{Y}$	Matrix of all multisource labels.
R_L	Matrix of labelwise reliability estimation.
R_I	Matrix of classwise probability.

I. INTRODUCTION

GROUND penetrating radar (GPR) is a commonly employed underground imaging technology that utilizes the transmission and reflection of electromagnetic waves to detect alterations in the dielectric properties of host materials [1], [2], [3]. This technique plays a pivotal role in evaluating the operational health of urban roads and subterranean facilities. Through the analysis of the shape or feature of B-scan images, researchers can effectively estimate the underground conditions or identify existing objects. This approach, interpreting subterranean situations based on image characteristic analysis, opens up an expansive application space for computer-aided technologies, including machine learning (ML) and deep learning (DL) methods [4].

Learning-based methods typically require a large-scale sample set with high-quality annotations [5]. Traditional data collection and learning model construction encounter substantial challenges when applied to GPR tasks. Specifically, the B-scan images generated by different underground conditions vary across diverse subterranean environments [6], [7], [8]. Even within the same type of underground condition, the features manifested in the GPR B-scan images display a degree of variation. A prime example is the cavity, in which the waveforms differ depending on their size, shape, medium environment, and fillers [9]. These complexities often constrain learning models to be trained primarily on images from limited detection areas in similar environments [10]. When these models are applied to an unfamiliar detection area, their effectiveness frequently diminishes due to differences in underground conditions. Consequently, for various underground detection tasks, organizing and labeling new GPR images are pivotal for accurate data recognition.

Manuscript received 16 January 2024; revised 15 April 2024, 13 June 2024, and 22 July 2024; accepted 13 August 2024. Date of publication 26 August 2024; date of current version 30 August 2024. This work was supported in part by the National Key Research and Development Program of China under Grant 2021ZD0111700; and in part by the National Nature Science Foundation of China under Grant 62206261, Grant 62137002, and Grant 62176245. (Corresponding authors: Xiangyu Wang; Huanhuan Chen.)

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Digital Object Identifier 10.1109/TGRS.2024.3446029

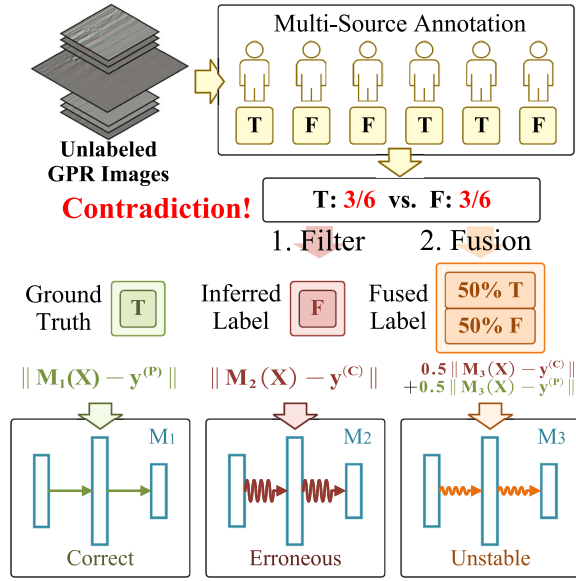


Fig. 1. Challenges when meeting great contradictions in the multisource annotation of GPR images. The filter solution may infer a wrong label and bring an erroneous DL model. The fusion solution maintains the possibility of annotation, which may still render the DL model unstable.

In this scenario, integrating multiple annotators to construct high-quality datasets becomes a practical way to improve task efficiency.¹ However, due to the professional level of annotators [11] and the difficulty of image identification, errors or contradictions inevitably exist in multisource labels [12], [13]. These contradictions may cause the model to fail to learn effective image features, especially when the samples confuse the annotators. To address these issues, some multisource aggregation-based methods are proposed to integrate diverse annotations during the learning process or design annotation filtering strategies to recommend a specific annotation [14], [15]. The former approach tends to fuse information provided by each annotator, despite potential inaccuracies in some labels. The latter directly assigns annotations to images using the proposed strategies, even if numerous opposing labels refute it. Unfortunately, both methods may pose risks to the learning process² when there is a significant contradiction in the multisource annotations (see Fig. 1).

When annotations for a specific GPR image are inconsistent, both accurate annotations and erroneous annotations may coexist within the given annotation set. Therefore, to train a more effective learning model, it is crucial to: 1) maximize the positive impact of correct annotations on model training and 2) mitigate the risks posed by conflicting annotations in cases where there is significant label confusion or inconsistency. Consequently, it is necessary to estimate the reliability of annotations from different annotators during the learning process and to devise a strategy that modulates the contributions of both reliable and risky annotations to the model.

¹These methods prevalently construct datasets through the way of redundant annotation, i.e., each image is assigned to multiple annotators for annotation (as illustrated in Section II).

²The learning process here refers to the learning of the model in single-label classification tasks, i.e., each image corresponds to only one correct category.

To tackle these issues, this article proposes a multisource label learning framework. A probabilistic multisource labeling aggregation (PMLA) module is developed to achieve reliability estimation of multisource annotations, where the reliability of annotation results for various images, attributed to each individual annotator, is iteratively assessed. The crucial design of the PMLA module is to leverage the consistency and contradiction between the annotators' labels for each image, instead of naively accepting the known professional level of the annotators. If a large disagreement arises among the annotators, the reliability estimation process relaxes the result to include multiple values, thus maximizing the impact of correct labels and reducing the risks of complete error labels. Moreover, a label-sensitive regularization (LSR) module is proposed to further mitigate the interference of erroneous labels on model learning. This module, which is sensitive to the prediction probability of labels aggregated from output reliability, calculates a confusion weight for each image and regularizes the training intensity. This is quite beneficial for avoiding the risk of contradictory annotations when a significant contradiction exists in annotation results.

The contributions of this article are summarized as follows.

- 1) This article proposes a novel multisource label learning framework tailored for underground object recognition, addressing the frequent need for data labeling in new environments.
- 2) The proposed method introduces the PMLA module, which uniquely estimates label quality and mitigates the risks associated with accumulated errors in single-label selection methods. This is further augmented by an LSR technique to align features more closely with ground truth.
- 3) By integrating label quality estimation directly into the learning pipeline and addressing the accumulated errors, our approach simplifies the overall process, reducing costs and enhancing the robustness of models for GPR-related tasks.

The rest of this article is organized as follows. Section II discusses the relevant backgrounds for the study. The proposed method is thoroughly explained in Section III. Section IV is dedicated to presenting the experimental outcomes and their subsequent analysis. The article is concluded in Section V.

II. RELATED WORK

As an important geophysical tool, GPR uses electromagnetic waves to produce detailed subsurface images, effectively uncovering complex structures, such as pipelines, cracks, and cavities [16]. These structures interact differently with the radar, forming intricate patterns that necessitate expert analysis for accurate interpretation. Due to the significant advantages of DL methods in automatically extracting and integrating multilevel features to model these complexities, there has been a recent surge in the development of DL-based approaches for automated complex feature extraction in GPR images [17], [18]. Several approaches have successfully built high-performance underground recognition models by utilizing B-scan images from specific underground areas as training data [19], [20]. For example, Kim et al. [21] collected data

from the underground of the city roads in Seoul and subsequently built a triplanar deep convolutional neural network for identifying underground such as cavities and pipelines. The advantage of these methods lies in their ability to conduct targeted analysis of the environmental features of specific underground areas to obtain high-performing models. Compared to traditional methods, they have significant advantages in both automation level and performance. However, the construction of such models usually depends on the high-quality annotation of specific underground data, making them difficult to apply directly to other tasks [22], [23].

Some studies in the field of GPR image analysis have also attempted to address this issue. Certain works try to use transfer learning strategies to reduce this dependence on data, intending to leverage pretrained models to accelerate model convergence and reduce the need for annotated training samples. For example, Li et al. [24] constructed a pretrained model using simulated data and fine-tuned the model using a small amount of annotated underground cavity data, with the aim of improving recognition accuracy in scenarios with limited annotated data. However, some researchers have pointed out that the high-level representations related to class membership in the pretrained dataset exhibit significant differences from the targets in the GPR data. Transfer learning strategies may only help DL models extract some low-level representations, such as lines and gray-scale values [25], [26]. Therefore, their practicality in different tasks faces challenges.

In response to this, many applications have adopted an alternative, simpler, and more practical method for constructing training data. This method typically involves distributing collected data among multiple annotators for labeling [27], aiming to quickly build large-scale, high-quality training data. The implementation process is simple and easy to apply. However, some researchers have pointed out that ensuring the annotation quality from multiple annotators is challenging, and the labels provided may include unreliable annotations [28]. To address this problem, a common practice is to mitigate the impact of erroneous labels by integrating redundant annotations. This typically involves repeatedly assigning the same sample to different annotators to enhance annotation fault tolerance. Based on this, how to integrate redundant annotations from multiple sources to build a more reliable model has become a focus of some studies.

Currently, based on the way multisource labels are aggregated, existing learning models can be mainly categorized into two types [29]. First, some methods design independent reliable label selection strategies to choose the most trustworthy label for each sample and then construct the model. This is essentially the same as single-label learning tasks. For this approach, constructing high-performance label selection strategies is key to improving model performance. Therefore, several multisource label selection strategies have been developed. For instance, Zhang and Wu [30] designed a model that infers the true labels for each sample by using a set of confusion matrices and the expectation-maximization algorithm. Zhang et al. [31] proposed an adaptive voting noise correction algorithm to accurately identify and correct potential erroneous labels and then used high-quality labels to

construct classification models. However, these methods may face the risk of error accumulation, i.e., the performance of the model is limited by the performance of the label selection strategy. When the selected label is incorrect, samples with erroneous labels can severely disrupt the learning process.

Second, another strategy optimizes the structure of the DL model and aggregates multisource labels during the training process to achieve label probability learning. For example, Rodrigues and Pereira [32] embedded an additional crowdsourcing layer after the output layer of the neural network to fuse the annotation information provided by different annotators. Li et al. [33] proposed a colabeling learning method to alternately train the image classification model and the label aggregation model. The advantage of these methods is that they consider the label information provided by different annotators, enabling the model to capture more effective features. However, when there is a large disagreement among different annotators, directly aggregating information provided by all annotators to guide model learning can introduce a large amount of erroneous label information and harm the model.

The motivation for this work arises from the pressing need to build high-quality, transferable DL models for underground object recognition, which are highly dependent on the precise annotation of data. Given the challenges in model transferability across different underground environments and the high demand for annotation accuracy, it is important to effectively integrate redundant annotations and ensure data quality. The proposed framework is designed to enhance the performance of models on such multisource data, balancing high annotation quality with flexibility for effective application across various subsurface conditions.

III. METHOD

In this section, we introduce the multisource label learning framework for classifying underground objects. The section begins with preparation for collected GPR data, followed by a formal definition of the multisource label learning task. Subsequently, an overall depiction of the framework's architecture is provided. The rest of the section provides a more granular explanation of the component details.

A. Preparation of GPR Data

To acquire GPR data with images that have prominent target features and accurate annotations, it is necessary to prepare the GPR data from collected GPR signals.

1) *GPR Data Preprocessing*: The GPR data are collected within the detection area, covering various subsurface conditions. The collected GPR data cannot directly display the appearance of subsurface objects [34]. The shape, type, and growing degree of the objects affect the reflected radar waves. It is essential to preprocess data to enhance the features of underground object anomaly signals before employing DL models [35].

The preprocessing of collected GPR data involves three steps: removing ground surface echo, reducing noise, and compensating for propagation losses. Initially, the reflectance

of the ground surface is eliminated in advance, which is facilitated by the MATGPR tool, a free MATLAB package designed specifically for the analysis of common-offset GPR data [36]. Following this, electromagnetic noise and interferences are mitigated using a standard median filter [37] to ensure the clarity of the data. Finally, the data undergo a compensation for the medium's attenuation and the radial dispersion of signal energy through the application of a nonlinear time-varying gain [38]. The goal of these steps is to eliminate noise and enhance the visibility of subsurface structures in the GPR data. Following these geophysics-based processes, experienced experts can identify objects of interest from the images.

2) *Data Annotation*: The preprocessed data undergo a thorough examination to identify and categorize various subsurface objects, with each object's region clearly delineated, including the specific start and end positions within the GPR images. Expert analysis is essential at this stage to ensure precision in the identification and localization process. These meticulously defined annotations serve as the gold standard or the ground truth for subsequent methodologies. To further validate the accuracy of these identifications, we conducted on-site verifications of selected objects. The step of verification guarantees that the annotations are reliable and can be effectively utilized for subsequent model training and validation.

3) *Image Segmentation*: Based on ground-truth annotations provided by experts, the preprocessed GPR B-scan trajectory images are segmented into multiple data frames. Regions containing identified objects are segmented with a fixed overlap, ensuring complete coverage of each object. Normal areas are randomly segmented from regions that do not contain any objects. Frames from regions with objects are then categorized according to the type of object that they contain. Notably, frames that include even a minimal extent of an object are labeled as the corresponding object type, as the detection of underground objects should ideally avoid any omissions.

In real-world scenarios, due to the high cost of expert resources, annotators are not always professionals. When annotations are of suboptimal quality, automatic recognition of underground objects remains a critical challenge. The rest of the sections will detail the methods to address this issue. To support such research, several volunteers from various levels of expertise, including experts, practitioners, and students, are engaged to perform repeated annotations on these images. This approach creates multisource annotations, which are used for model learning in expert-lacking scenarios.

B. Problem Definition

The input of GPR B-scan images is denoted as $\mathbf{X} = \mathbf{X}^k \cup \mathbf{X}^u$, where $\mathbf{X}^k$ represents images with known ground truth and $\mathbf{X}^u$ represents images without known ground truth. $\mathbf{X}^u$ contains m images, which is much more than the image amount t in $\mathbf{X}^k$. Let x_j^u denote the j th image in $\mathbf{X}^u$, which is labeled by n annotators $\mathcal{A} = \{a_1, a_2, \dots, a_n\}$ using preset GPR object class $\mathcal{C} = \{c_1, c_2, \dots, c_r\}$, where r is the total number of object classes. The annotation results of x_j^u in $\mathbf{X}^u$ are denoted as $\mathbf{u}_j = (u_{1,j}, u_{2,j}, \dots, u_{n,j})^T$, where $u_{i,j} \in \mathcal{C}, i = 1, 2, \dots, n$. Similarly, x_j^k is the j th image in $\mathbf{X}^k$, having multisource label

$\mathbf{k}_j = (k_{1,j}, k_{2,j}, \dots, k_{n,j})^T, j = 1, 2, \dots, t$. The ground truth of $\mathbf{X}^k$ is represented as $\mathbf{GT}^k = \{g_1^k, g_2^k, \dots, g_t^k\}$. Supposing that $\mathbf{Y} = (\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m, \mathbf{k}_1, \mathbf{k}_2, \dots, \mathbf{k}_t)$ is a matrix containing all multisource labels, the goal of the article is to infer the most reliable annotations for the images $\mathbf{X}^u$ using images $\mathbf{X}^k$ and multisource labels $\mathbf{Y}$, further training an effective model M on both $\mathbf{X}^k$ and $\mathbf{X}^u$.

Note that when representing any annotator in $\mathcal{A}$, only the variable subscript i is used so that readers can easily judge the meaning of the symbols in the following formulas. Similarly, only subscript j represents an image in $\mathbf{X}^k$ or $\mathbf{X}^u$, and only subscript s represents an object class in $\mathcal{C}$. The descriptions of symbols used in the method are summarized in Nomenclature.

C. Overall Framework

The proposed framework consists of two main modules, the PMLA module and the LSR module, to assist high-quality learning from multisource labels, as shown in Fig. 2.

PMLA generates a probability for each image in $\mathbf{X}^u$ as the learning target for predicting results $O = M(\mathbf{X}^u)$, which is obtained by feeding training images $\mathbf{X}^u$ to a DL model. Concretely, PMLA produces labelwise reliability estimation $R_L \in \mathbb{R}^{|\mathcal{A}| \times |\mathbf{X}^u|}$ using multisource labels $\mathbf{Y}$ and gather the results into a classwise probability matrix $R_I \in \mathbb{R}^{|\mathcal{C}| \times |\mathbf{X}^u|}$. Such images are trained with multiple labels through the matrix R_I rather than a single label to reduce the risk of erroneous label inferring.

On this basis, the LSR module limits the confused multilabel in the training process. The LSR module calculates imagewise regularization weights vector W , a component of which is large when corresponding images have similar probability across different classes. Combined regularization weights, the proposed method is biased toward greatly reliable multilabel and adopts low-weighted multilabel training on images with confused estimated reliability to prevent misleading by wrong annotation.

D. Probabilistic Multisource Label Aggregation

In the PMLA module, a limited number of samples with ground-truth labels are first utilized to roughly estimate each annotator's professional level and initialize annotation reliability. Then, using a larger number of multisource annotations, PMLA calculates the consistency of each annotation to adjust the initial reliability estimates, continuously improving the estimation accuracy of label reliability.

The PMLA module produces a probabilistic matrix with input $\mathbf{Y}$, denoted as $R_I = A(\mathbf{Y})$. The overall equation is presented as follows:

$$A(\mathbf{Y}) = \text{Softmax}(T(\mathbf{Y}, E(\mathbf{Y}))) \quad (1)$$

where $E(\cdot)$ is the reliability estimation module for multiple labels on each GPR B-scan image, $T(\cdot, \cdot)$ gathers the estimated reliability to produce reliability evaluation on the B-scan image class, and Softmax normalizes $T(\mathbf{Y}, E(\mathbf{Y}))$ such that the output $A(\mathbf{Y})$ is a vector of probabilities that sum to 1.

$E(\cdot)$ simulates the process of multiple rounds of debate, where annotators with different reliability debate the true label

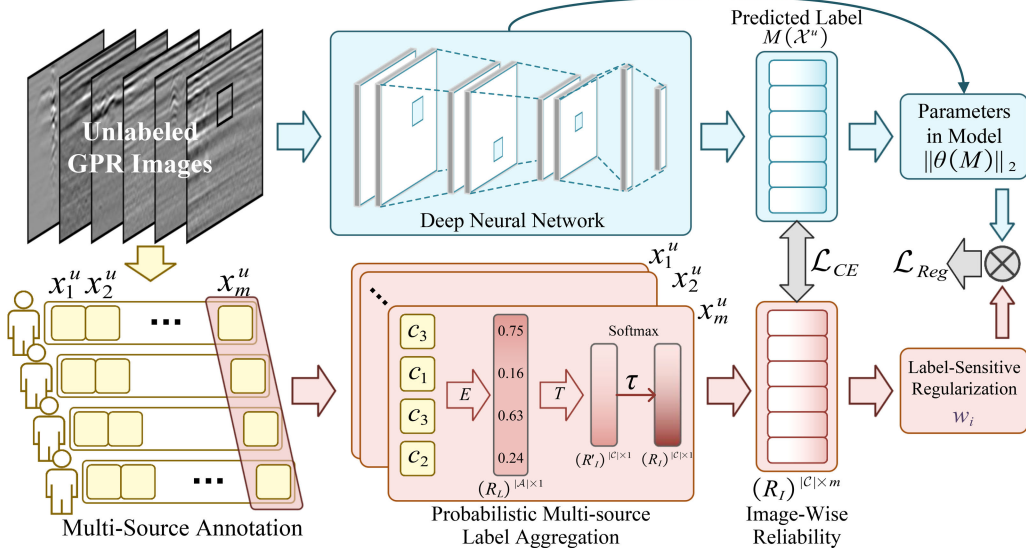


Fig. 2. Illustration of the framework of the proposed method. The PMLA model is designed to infer the quality of each annotation, evaluate the probability of each class, and aggregate multisource annotations into imagewise reliability. In addition, the label-sensitive w_i are calculated from imagewise reliability to control the weight of the regularization term. The symbol $(\cdot)^{\times}$ represents the shape of the matrix, where $|A|$ and $|C|$ represent the numbers of annotators and B-scan class, respectively. $\mathcal{L}_{CE}$ and $\mathcal{L}_{Reg}$ are two losses introduced in Section III in detail.

of an image. The reliability of annotators who are favored will increase, while annotators who are widely negative will not be trusted. More formally, the professional level of annotator $P = \{p_1, p_2, \dots, p_n\}$ is first calculated according to their labeling result on $\mathbf{X}^k$, and an iterative process based on P is designed to estimate the reliability of multisource labels on $\mathbf{X}^u$. Let $\mu_{i,j}^0$ represent the initial reliability of label $u_{i,j}$, initialized as

$$\mu_{i,j}^0 \leftarrow p_i \quad i \in \{1, 2, \dots, n\}, \quad j \in \{1, 2, \dots, m\} \quad (2)$$

where p_i is the professional level of annotator a_i , calculated by the label accuracy of annotators on $\mathbf{X}^k$

$$p_i = \frac{\sum_j \mathbb{I}(k_{i,j} = g t_j^k)}{t} \quad (3)$$

where $\mathbb{I}(\cdot)$ is the indicator function that values 1 when the statement in the brackets holds and values 0 when not.

The reliability of labels on each image is updated iteratively through a process incorporating the consistency and reliability of the previous round, denoted as

$$\mu_{i,j}^{l+1} = \mu_{i,j}^l + \sum_{e \neq i} B(\mu_{i,j}^l, u_{i,j}, u_{e,j}) \times U(\mu_{i,j}^l, \mu_{e,j}^l) \quad (4)$$

where $\mu_{i,j}^l$ is the reliability of $u_{i,j}$ after the l th iteration and $u_{e,j}$ is the result from annotator a_e that is being compared with $u_{i,j}$. $B(\cdot)$ is the upper bound of the update, and $U(\cdot)$ denotes the update amount. $B(\cdot)$ is designed as follows:

$$B(\mu, u_1, u_2) = \mathbb{I}(u_1 = u_2) \times B_u(\mu) - \mathbb{I}(u_1 \neq u_2) \times B_d(\mu) \quad (5)$$

where u_1 and u_2 are the two currently compared annotations and μ is the reliability of u_1 . $B_u(\mu)$ is the upper bound for increasing reliability, and $B_d(\mu)$ is the upper bound for decreasing reliability. This article uses the following formula to calculate $B_u(\mu)$ and $B_d(\mu)$:

$$B_u(\mu) = 1 - \mu, \quad B_d(\mu) = \mu. \quad (6)$$

The update amount $B(\mu, u_1, u_2)$ limits the updated reliability in the range of $(0, 1)$. Moreover, it indicates the update direction of the reliability, which increased when agreed by other annotators and decreased when not.

The update amount $U(\cdot)$ in (4) is designed as

$$U(\mu_1, \mu_2) = \mu_2 \times (\mu_2 - \mu_1 + 1). \quad (7)$$

Intuitively, the update bound is determined by two terms: the former term represents the absolute reliability of the compared label $\mu_{e,j}$, and the latter term represents its relative gap to the investigated label reliability $\mu_{i,j}$. This indicates that more reliable labels matter more to modifying other labels' reliability, and estimation on less reliable labels is more easily affected by other labels, and vice versa. The designed reliability estimation procedure places more emphasis on the annotator prior and can better handle the case that truth presents in the few.

Through the estimation process $E(\cdot)$ with an iteration limit l_0 , we obtain the estimated labelwise reliability matrix

$$E(\mathbf{Y}) \leftarrow R_L = (\mu_{i,j}^{l_0})^{n \times m}. \quad (8)$$

On this basis, we perform a weighted voting operation $T(\cdot, \cdot)$, which aggregates the results of different annotators to obtain an imagewise reliability matrix on different labels

$$T(\mathbf{Y}, R_L) \leftarrow R'_l = (q_{s,j})^{r \times m} \quad (9)$$

where the detailed transformation is presented as

$$q_{s,j} = \sum_{a=1}^n \mathbb{I}(u_{a,j} = c_s) \times \mu_{a,j}^{l_0} \quad (10)$$

where c_s is the s th class. The imagewise reliability matrix R_l is subsequently processed with a softmax function

$$R_l = \text{Softmax}(R'_l) = [\sigma(\mathbf{q}_1) \sigma(\mathbf{q}_2), \dots, \sigma(\mathbf{q}_m)] \quad (11)$$

where $\mathbf{q}_j = (q_{1,j}, q_{2,j}, \dots, q_{r,j})^T$ and $\sigma(\cdot)$ is the softmax function that transfers the reliability to probability

$$\sigma(\mathbf{q}_j) = \left[\frac{\exp(\tau q_{1,j})}{\sum_{s=1}^r \exp(\tau q_{s,j})}, \dots, \frac{\exp(\tau q_{r,j})}{\sum_{s=1}^r \exp(\tau q_{s,j})} \right]^T \quad (12)$$

where temperature parameter τ is a positive real number, which can adjust the distribution of the softmax function. τ being close to 0 will make the probabilities of different image categories have a larger gap, and τ being larger means that the output probability distribution of the softmax function will be more even. $(R_l)_{s,j}$ indicates the estimated probability that the j th image in $\mathbf{X}^u$ belongs to class c_s , which is used to supervise the training process

$$\begin{aligned} \mathcal{L}_j &= \text{CrossEntropy}(R_l, M(\mathbf{X}^u); j) \\ &= - \sum_{s=1}^r (R_l)_{s,j} \log M(x_j^u)_s \end{aligned} \quad (13)$$

where $M(x_j^u)_s$ represents the probability of c_s predicted by model M for x_j^u . The total loss is

$$\mathcal{L}_{\text{CE}} = \frac{1}{m} \sum_{j=1}^m \mathcal{L}_j. \quad (14)$$

The physical interpretation of PMLA lies in its iterative process where, in each round, PMLA calculates the update bounds for the quality of each label based on the pairwise consistency of multisource labels. Using the quality estimates from the previous round and the multisource labels, it automatically computes the update amount for each label's quality. Thus, the update bounds are determined by the quality itself, while the update amount results from a multisource label voting process that incorporates quality factors. The estimation method designed in PMLA ensures that more reliable labels play a more significant role in updating other labels. The combination of update bounds and update amount in PMLA ensures that each iteration optimizes the update of multisource label quality in the correct direction and with the optimal step size, minimizing the risk and making the estimate increasingly close to the true value. This enables accurate probabilistic estimation of the confusable physical characteristics of underground objects in GPR. The process of PMLA is formally summarized in Algorithms 1 and 2.

The framework typically assigns an image label based on the class with the highest estimated reliability. However, if multiple classes have similar estimated reliability, the image is trained with multiple labels, making reliability untrustworthy for indicating correct labels. In such cases, R_l is used to calculate imagewise weights in the LSR, as introduced in Section III-E.

E. Label-Sensitive Regularization

The LSR module is conducted to alleviate the effect of untrustworthy estimation. The imagewise weights in the regularization process are calculated by

$$V(R_l) \leftarrow W = (w_j)^{m \times 1} \quad (15)$$

Algorithm 1 E: Reliability Estimation in PMLA

Input: Multisource labels $\mathbf{Y}$; The ground truth GT^k of $\mathbf{X}^k$
Output: Estimated reliability R_L .

```

1 for  $i=1:n$  do
2    $p_i = \frac{\sum_j \mathbb{I}(k_{i,j}=gt_j^k)}{t}$ ; // Prior calculation
3 end
4 for  $i=1:n, j=1:m$  do
5    $\mu_{i,j}^0 = p_i$ ; // Initialization
6 end
7 for  $l=0:l_0-1$  do
8   for  $i=1:n, j=1:m$  do
9      $B_e = \mathbb{I}(u_{e,j} = u_{i,j}) \times (1 - \mu_{i,j}^l) - \mathbb{I}(u_{e,j} \neq$ 
10       $u_{i,j}) \times \mu_{i,j}^l$ ; // Equation (5)
11      $U_e = (\mu_{e,j}^l - \mu_{i,j}^l + 1) \times \mu_{e,j}^l$ ;
12     // Equation (7)
13      $\mu_{i,j}^{l+1} = \mu_{i,j}^l + \sum_{e \neq i} B_e \times U_e$ ;
14     // Equation (4)
15   end
16 end
17  $R_L \leftarrow (\mu_{i,j}^{l_0})^{n \times m}$ ;
18 Return  $R_L$ ;
```

Algorithm 2 Probabilistic Multisource Label Aggregation

Input: Multisource labels $\mathbf{Y}$; The ground truth GT^k of $\mathbf{X}^k$
Output: Probability matrix R_l .

```

1  $R_L = E(Y)$ ;
2 for  $s=1:r, j=1:m$  do
3    $q_{s,j} = \sum_{a=1}^n \mathbb{I}(u_{a,j} = c_s) \times (R_L)_{a,j}$ ;
4   // Equation (10)
5 end
6 for  $s=1:r, j=1:m$  do
7    $q'_{s,j} = \frac{\exp(q_{s,j})}{\sum_{s=1}^r \exp(q_{s,j})}$ ; // Equation (12)
8 end
9  $R_l \leftarrow (q'_{s,j})^{r \times m}$ ;
10 Return  $R_l$ ;
```

where the calculation of w_i is presented as follows:

$$w_j = \alpha \times \left(1 - \sum_s (R_l)_{s,j}^2 \right) \quad (16)$$

where α is a hyperparameter and $(R_l)_{s,j}$ is the probability that the j th image belongs to class c_s . w_j actually indicates the extent of confusion of the reliability estimation on the j th image, whose characteristic is briefly illustrated as follows.

Equations (11) and (12) suggest that $\sum_s (R_l)_{s,j} = 1$ and $(R_l)_{s,j} \geq 0$. Consider various cases of valuing.

- 1) *Maximum case:* We have $\sum_s (R_l)_{s,j}^2 \geq ((\sum_s (R_l)_{s,j})^2 / r) = (1/r)$, where value $(1/r)$ is taken iff $\forall s, (R_l)_{s,j} = (1/r)$. That is, all the classes are estimated with identical probability as the true

label, i.e., the estimation process completely fails and provides no effective information for label selection at all. w_j takes the maximum value $\alpha \times (r - 1/r)$ in this case.

- 2) *Minimum case*: We have $\sum_s (R_I)_{s,j}^2 \leq (\sum_s (R_I)_{s,j})^2 = 1$, where value 1 is taken iff $\exists s, (R_I)_{s,j} = 1$. That is, the estimation process is fully confident with the produced true label. w_j takes the minimum value 0 in this case.
- 3) *Middle cases*: Suppose that r_0 classes are assigned with identical probability $(1/r_0)$, i.e., the estimation results indicate that one of the r_0 classes is the true label but completely unknown of which, and then, $w_j = \alpha \times (r_0 - 1/r_0)$. In this case, w_j is larger with larger r_0 , which means that the estimation is confused by more classes, i.e., the estimated results have less effective information for indicating the true label.

The analysis above shows what the confusion extent represented by w_j is. In a word, the estimation is less confusing if a larger probability are assigned to fewer classes, and it is more confusing when many classes are estimated with close probability.

w_j in (16) makes further restriction on the loss function in (13) in a following way:

$$\mathcal{L}_{\text{Reg}} = \sum_j w_j \|\theta(M)\|_2 \quad (17)$$

where $\theta(M)$ represents the vector of parameters of the image classifier and $\|\cdot\|_2$ is the L^2 -norm.

Finally, the loss function of the total model is concluded as

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{Reg}} \quad (18)$$

where $\mathcal{L}$ is the final loss function for the DL model to minimize.

IV. EXPERIMENTAL RESULTS

A. Datasets

The utilized GPR B-scan image collections are GPR-SD,³ Road [28], and DECKGPRH [39].

1) *GPR-SD*: GPR Subsurface Diagnosis (GPR-SD) uses GPR to collect images from concrete and asphalt roads at frequencies between 200 and 400 MHz. Three parallel GPR B-scan images are taken with a 3-cm trace interval containing 380 samples in a 97-ns time window. The MATGPR tool [36] processes the data, removing ground echoes and applying gain for subsurface targets. Five target types are identified and marked, and the images are cropped, resulting in 930 scenes of 321×321 pixels each. Example images are displayed in Fig. 3.

2) *Road*: The dataset comprises GPR B-scan images collected from three different road surfaces: concrete, asphalt, and unpaved. These images are captured using the GSSI SIR-30 GPR system, integrated with GPS technology, operating at frequencies between 50 and 600 MHz. The system records data with a time window of 70 ns and samples 512 data points, with a trace interval of 2 cm. To improve data quality, these images undergo various processing techniques, including

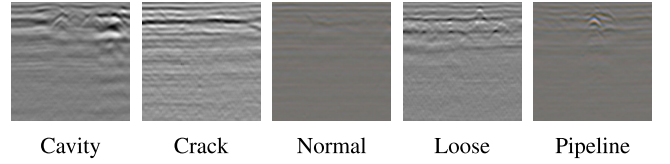


Fig. 3. Images sampled from the GPR-SD dataset.

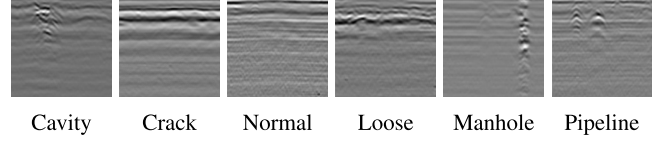


Fig. 4. Images sampled from the Road dataset.

TABLE I
DETAILS AND IMAGE COUNTS FOR EACH UNDERGROUND OBJECT CLASS IN THE GPR-SD AND ROAD DATASETS

Class	Description	GPR-SD	Road
Cavity	Air-filled underground structures of irregular shapes beneath roads, causing subsidence. Characterized by strong reflected EM waves and potential multiple reflections and diffraction due to air's lower dielectric constant.	180	154
Crack	Presence of horizontal gaps within the compact underground road structure, typically filled with air. These cracks cause wave reflections and abrupt discontinuities in waveforms.	150	189
Normal	Areas without any subsurface anomalies or objects, indicating a standard, undisturbed underground condition.	200	147
Loose	Soil areas with higher porosity and lower density, resulting in reduced cohesion and compactness. This condition, akin to a mix of soil and air, leads to chaotic reflected waves in GPR data due to the irregular arrangement of soil properties.	230	161
Manhole	Presence of a manhole structure beneath the road surface.	-	153
Pipeline	Presence of a pipeline object beneath the road surface.	170	201

removing surface echoes, reducing noise, and compensating for propagation loss. The dataset identifies and labels six distinct types of subsurface targets, rigorously tested and manually adjusted in the field for accuracy; 1005 images are cropped from this collection, each with a resolution of 321×321 pixels. Example images are shown in Fig. 4.

To enhance our examination of class distinctions within the GPR-SD and Road datasets, Table I offers a comprehensive breakdown of the unique characteristics of each class. This includes an in-depth description of distinctive features and a tally of sample counts across both datasets. For concise visualization of the relationships among classes, vector representations are computed for each class using the pretrained Momentum Contrast (MoCov2) model [40]. The similarity matrix is constructed by calculating the cosine similarity between vectors for each pair of classes. Specifically, for the k th image of class C_i , the pretrained encoder extracts the vector representation v_i^k . The average representation $\bar{v}_i$ for each class C_i is then calculated by averaging the vectors of all m images in that class: $\bar{v}_i = (1/m) \sum_{k=1}^m v_i^k$. The cosine similarity between two classes, C_i and C_j , is given by

$$\text{CosSim}(C_i, C_j) = \frac{\bar{v}_i \cdot \bar{v}_j}{\|\bar{v}_i\| \|\bar{v}_j\|} \quad (19)$$

where $\|\bar{v}_i\|$ and $\|\bar{v}_j\|$ represent the magnitudes of the vectors $\bar{v}_i$ and $\bar{v}_j$, respectively. Finally, the similarity matrices are constructed and presented in Fig. 5.

³Resources could be obtained from https://github.com/ubriustc/GPR_data.

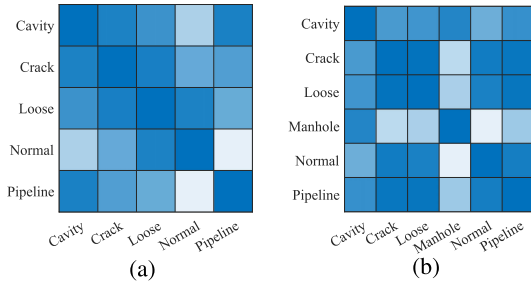


Fig. 5. Cosine similarity matrices among classes. (a) GPR-SD. (b) Road.

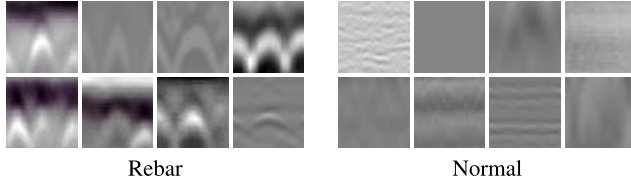


Fig. 6. Images sampled from the DECKGPRH dataset.

3) *DECKGPRH*: The DECKGPRH dataset is proposed by [39], containing data with two classes of images: Rebar and Normal. The images are collected with a GSSI SIR System 3000 GPR, using a 1.6-GHz antenna, and are cart-mounted for scanning concrete bridge decks. Each Rebar image in the dataset features a hyperbola, representing the detection of rebar objects, while the normal samples are hyperbola-free images. This study uses a balanced dataset of 1600 images for the experiment, comprising an equal number of both classes. Example images are shown in Fig. 6.

Seven real-world datasets with multisource labels are constructed, including three with real annotations: (A) GPR-SD, (B) Road, and (C) DECKGPRH, each annotated by six real volunteers, and four with simulated annotations: (D) Road-G1, (E) Road-G2, (F) Road-G3, and (G) Road-G4, each annotated by five simulated annotators. The details of the simulated annotation process are discussed in Section IV-B1.

All datasets are split with an 80% allocation for model training and 20% for model testing. The training data comprise two subsets: 30% of images with known ground truths, denoted as the validation (Valid) set, and 70% with unknown ground truths, denoted as the training (Train) set. The validation set is utilized for initial reliability in PMLA and model evaluation, whereas the training set is employed for the training of PMLA and neural networks. The testing data are denoted as the Test set. Each object class maintains the same proportion in these divisions.

B. Experimental Setup

1) *Details of the Simulated Annotation Process*: To comprehensively study and evaluate the proposed methods' stability across various scenarios, four simulated annotation datasets, (D) Road-G1, (E) Road-G2, (F) Road-G3, and (G) Road-G4, annotated by five simulated annotators, are built based on the real-world GPR images in the Road dataset.

In these simulated annotation datasets, each annotator is set to label an image correctly with a preset probability. Formally

TABLE II
SETTINGS OF PROFESSIONAL LEVELS OF ANNOTATORS
IN SIMULATION DATASETS

Dataset	A_1	A_2	A_3	A_4	A_5	Avg	Std
Road-G1	0.95	0.80	0.60	0.50	0.25	0.62	0.27
Road-G2	0.80	0.70	0.60	0.40	0.30	0.56	0.21
Road-G3	0.85	0.60	0.50	0.30	0.20	0.49	0.26
Road-G4	0.94	0.94	0.92	0.92	0.90	0.90	0.02

speaking, for each GPR image in a simulated dataset with r categories, the i th simulated annotator A_i will correctly annotate the image with a set professional level p_i and randomly annotate the image as each incorrect class with the probability $(1 - p_i / r - 1)$.

The quantitative descriptions of these datasets are displayed in Table II. The columns labeled A_1 – A_5 indicate the professional levels of the simulated annotators, with both the average accuracy and the standard deviation calculated accordingly. In these simulated datasets, the levels of professional expertise vary: the average skill levels of Road-G1/4 are 0.62, 0.56, 0.49, and 0.90, respectively, representing different annotation scenarios.

2) *Parameter Settings*: Striking a balance between retaining the essential features of the GPR images and accelerating the learning process, we standardized the images to a resolution of 128×128 pixels before feeding them into the neural network in batches of 64. Models are trained using the Adam optimizer with a learning rate of 0.001 for 500 epochs. To ensure rapid initial progress while gradually reducing the rate to stabilize training and fine-tune the model's convergence toward the optimal solution, an exponential decay strategy is employed for the learning rate with a decay coefficient of 0.999.

During training, we evaluate accuracy after each epoch and select the model with the highest validation accuracy as the final model, ensuring optimal performance and minimizing overfitting. All experiments are conducted on a dedicated server, equipped with Ubuntu 20.04.5 LTS as the OS, a 12th-Gen Intel Core i5-12400 CPU, and an Nvidia GeForce RTX 3090 GPU.

3) *Evaluation Metrics*: In evaluating the models, accuracy is the main metric, defined as the ratio of correct predictions to total instances. The balanced sample distribution across classes in our dataset, as shown in Table I, justifies the use of accuracy for reliable performance measurement.

To assess individual module performance in this task, we use two accuracy metrics: label accuracy and learning accuracy. Label accuracy measures the accuracy of annotated or algorithm-aggregated labels against ground truth. Learning accuracy evaluates the DL model's Test set accuracy, trained with either multisource or single-source labels. The dual approach provides a comprehensive module performance analysis.

C. Results and Analysis

1) *Benefits From Multisource Label Learning*: To demonstrate the efficacy of multisource label learning, the proposed multisource label learning framework is compared with the

TABLE III
LABEL AND LEARNING ACCURACY ON GPR-SD

Aggregation Method	Label Accuracy	Learning Accuracy			
		MobileNet	VGG	ResNet	DenseNet
A_1	57.17	52.88 $\pm$ 3.29	55.62 $\pm$ 3.67	56.60 $\pm$ 2.32	56.73 $\pm$ 3.23
A_2	63.31	48.98 $\pm$ 1.98	54.31 $\pm$ 4.43	59.90 $\pm$ 2.42	58.58 $\pm$ 3.97
A_3	27.87	24.42 $\pm$ 2.28	25.06 $\pm$ 3.63	27.24 $\pm$ 1.60	27.59 $\pm$ 2.65
A_4	53.70	40.61 $\pm$ 1.42	44.79 $\pm$ 2.74	48.18 $\pm$ 2.45	47.28 $\pm$ 1.87
A_5	92.28	84.72 $\pm$ 1.90	84.15 $\pm$ 2.79	88.59 $\pm$ 1.31	87.41 $\pm$ 0.59
A_6	44.72	31.28 $\pm$ 2.20	37.45 $\pm$ 2.94	37.59 $\pm$ 2.73	37.23 $\pm$ 3.34
GLAD	82.83	81.03 $\pm$ 2.05	80.61 $\pm$ 1.34	82.26 $\pm$ 1.07	81.85 $\pm$ 1.38
RY	89.76	83.88 $\pm$ 2.73	83.44 $\pm$ 1.45	87.05 $\pm$ 1.29	87.39 $\pm$ 1.65
EM	83.31	78.67 $\pm$ 1.68	78.61 $\pm$ 1.73	82.87 $\pm$ 1.11	79.69 $\pm$ 1.76
PMLA	93.23	84.07 $\pm$ 3.10	88.82 $\pm$ 2.40	89.72 $\pm$ 1.26	88.74 $\pm$ 2.68
Ours	93.23	84.92$\pm$2.26	89.43$\pm$2.64	91.96$\pm$1.32	91.28$\pm$1.05
TRUE	100.0	88.51$\pm$4.73	89.67$\pm$2.29	92.36$\pm$0.83	92.61$\pm$0.59
Mem (MiB)	-	5.486	6.662	2.508	10.916
Time (s)	-	1,054.13	1,879.92	688.56	1,356.78

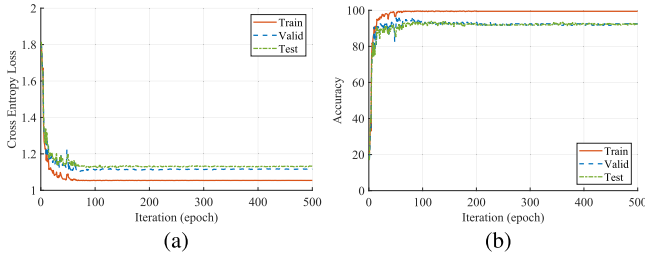


Fig. 7. Loss and accuracy metrics throughout the convergence process during the training of the DNN module. (a) Cross-entropy loss. (b) Accuracy.

single-label learning methods, which simply rely on annotations from single annotators (A_1 – A_6). Moreover, the proposed method is compared with three other common multilabel aggregation methods, including GLAD [41], RY [42], and EM [43], which use aggregate multisource labels into a single label and then feed into the DL model for training.

The performance of the proposed framework, along with single-label learning and other label aggregation learning methods, is evaluated using three real-labeled datasets [(A), (B), and (C)], and four DL models (MobileNetv2 [44], VGG-16 [45], ResNet-18 [46], and DenseNet-121 [47]). Experimental results are shown in Tables III–V. Experiments with models trained directly on true labels are also conducted (shown as TRUE), reflecting the difficulty of learning the dataset. Memory usage (Mem) and training time (Time) are recorded. Accuracy is highlighted when highest. The loss and accuracy metrics during the training process of the DNN module of dataset Road are shown in Fig. 7. Note that the reported accuracy of “PMLA” is approximated by binarizing probability labels. “PMLA” uses only the PMLA module, while “Ours” also employs the LSR module. As the LSR module does not affect label aggregation, “Ours” and “PMLA” share the same label accuracy.

Consider the experimental results obtained using the GPR-SD dataset as an illustrative example. Within this dataset, label accuracies span a broad spectrum, ranging from a modest 30% to an impressive 92%. Previous multilabel aggregation methods struggle to achieve a 90% threshold. In stark contrast, the proposed PMLA method consistently surpasses this mark, achieving a label accuracy in excess of 93%. This remarkable performance not only outshines the results of single-label learning but also eclipses those of multiannotation

aggregation methods. Such findings underscore PMLA’s exceptional capability in extracting latent high-quality labels.

Experiments suggest that an increase in label accuracy generally leads to higher learning accuracy, with a few exceptions. Hence, the proposed method’s strong performance in label accuracy often results in significantly superior learning accuracy compared to other methods. Notably, even without extreme label accuracy, the learning accuracy of the proposed method approaches the accuracy using ground-truth labels. This phenomenon suggests that the proposed method can leverage the available information from multisource labels effectively to approach the optimal result.

Among the utilized DL models, ResNet consistently outperforms the other model as a bottleneck. ResNet also proves to be the most efficient, with the least memory use and shortest training time. The ResNet-18 model, an 18-layer network, starts with a 7×7 convolutional layer (64 filters), batch normalization, ReLU activation, and 3×3 max pooling. It features eight residual blocks, each with two convolutional layers (64, 128, 256, and 512 filters), batch normalization, and ReLU activation, with downsampling in the first block of each set. Identity shortcuts in each block address the vanishing gradient problem. The network concludes with global average pooling and a fully connected layer, outputting class probabilities. For brevity, only the results using ResNet will be presented in future comparison and ablation experiments.

2) *Efficiency Analysis*: To verify the efficiency of the proposed method, a detailed image case study is provided. The cosine similarity matrices among classes are calculated in the same manner as the matrices shown in Fig. 5(b). As shown in Fig. 8, due to the similarity in features between “Loose” and “Crack,” many nonprofessional annotators might get confused. In this case, previous methods typically perform weighted voting based directly on the initially estimated reliability. However, this approach can be influenced by the predominance of low-quality labels, leading to an incorrect “Crack” label, which ultimately results in biased model training. In contrast, the PMLA method accurately computes the quality of different annotations, thus determining the weight of each annotation on the aggregated probabilistic label. Utilizing these accurate probabilistic labels for training and the LSR module’s assessment of overall reliability, the model is enabled to assimilate the correct information.

Reliability estimation significantly impacts PMLA, as shown in Fig. 9, which displays imagewise reliability after 20 iterations of the reliability estimation process of PMLA, for a subset of images. The figure shows annotation results for 16 GPR images from five annotators, with darker colors indicating higher reliability. The upper axis labels represent the PMLA label aggregation results, while the lower axis labels denote the ground-truth labels. After 20 iterations, correct annotations usually have higher reliability, while most of the lower reliability annotations are incorrect. Notably, our proposed method can identify the correct annotation when there is disagreement among annotators based on calculated reliability (as shown in Column 6). This highlights the PMLA module’s effectiveness in accurately filtering out correct annotations.

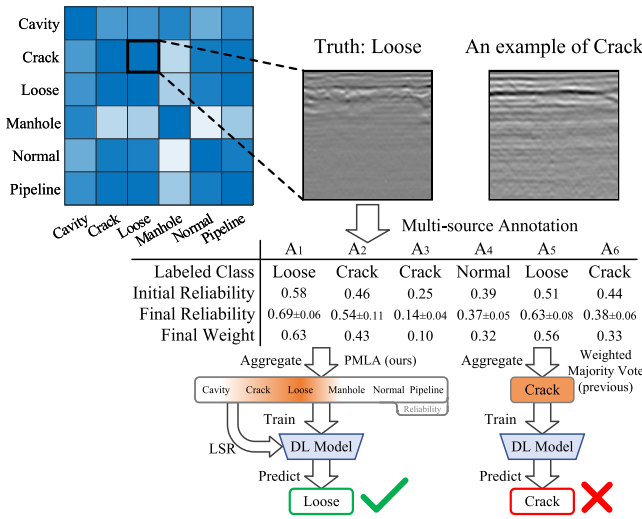


Fig. 8. Practical case of learning with multisource label.

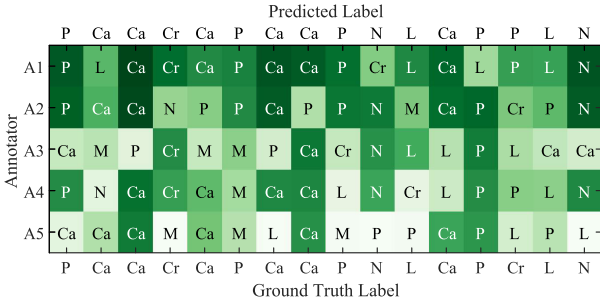


Fig. 9. Part of imagewise reliability after 20 iterations of the reliability estimation process of PMLA. The intensity of the background color corresponds to the quality of the label obtained by the PMLA model. The labels of P, V, Ca, Cr, L, and N are, respectively, represented for pipeline, manhole, cavity, crack, loose, and normal.

3) *Influence of the Professional Level of Annotators*: Different professional levels of annotators directly influence the quality of the label, which may further cause the performance gap in the learning model. In this section, the aggregated label quality and the final performance of the learning model on different annotator professional levels are tested, as shown in Table VI. Similar to Table IV, the columns “Ours” and “PMLA” use the same multisource label aggregation process and, therefore, have the same label accuracy. The “Label” and “Learning” columns represent the label accuracy and learning accuracy, respectively.

Fig. 10 illustrates the comparative performance between learning with single annotation (Single) and the proposed multisource label learning framework (Ours) across seven datasets. According to the results, traditional methods experience considerable variance and generally poor learning outcomes due to significant differences in annotation levels. In contrast, the proposed multisource learning framework effectively capitalizes on the strengths of all multisource labels, achieving significantly better results than traditional approaches. The findings underscore the considerable advantages of using multisource label learning, even when individual labels are highly accurate.

The result illustrates that the PMLA method has greatly surpassed other label aggregation methods in the label

TABLE IV
LABEL AND LEARNING ACCURACY ON ROAD

Aggregation Method	Label Accuracy	Learning Accuracy			
		MobileNet	VGG	ResNet	DenseNet
A ₁	94.06	86.47±2.60	87.27±2.27	89.06±1.26	90.34±1.98
A ₂	51.30	43.53±2.60	49.51±3.08	49.83±2.11	50.42±1.51
A ₃	30.14	26.32±1.36	28.07±3.75	25.64±3.66	25.47±3.51
A ₄	61.16	55.98±2.74	58.41±2.47	57.35±3.60	60.62±1.87
A ₅	72.75	64.10±4.14	69.56±3.58	67.69±3.74	69.06±2.75
A ₆	58.70	50.43±1.94	54.41±1.73	56.16±2.42	53.25±1.46
GLAD	90.87	80.50±2.94	84.31±2.29	86.63±1.94	87.38±2.54
RY	92.82	83.18±2.10	87.54±2.67	89.52±2.66	90.13±1.75
EM	91.30	81.48±2.36	83.41±2.38	86.39±2.11	87.52±0.59
PMLA	93.75	84.40±3.90	89.85±3.31	91.26±1.99	91.28±1.42
Ours	93.75	89.81±2.44	90.08±2.03	92.51±1.67	93.16±0.78
TRUE	100.0	90.26±1.95	91.14±1.80	94.95±1.27	94.98±1.48
Mem (MiB)	-	5,486	8,972	2,476	10,924
Time (s)	-	751.35	1,543.56	646.40	1,880.73

TABLE V
LABEL AND LEARNING ACCURACY ON DECKGPRH

Aggregation Method	Label Accuracy	Learning Accuracy			
		MobileNet	VGG	ResNet	DenseNet
A ₁	80.23	78.98±2.10	76.63±1.59	80.32±1.31	79.60±1.12
A ₂	86.54	83.23±2.21	67.24±1.41	83.50±1.93	83.07±2.24
A ₃	71.66	70.83±1.46	69.14±2.12	71.20±1.73	71.03±2.09
A ₄	61.32	60.22±1.81	54.98±2.31	60.52±2.26	57.22±1.10
A ₅	94.51	91.67±1.27	89.26±1.51	92.32±1.05	90.78±1.72
A ₆	74.76	74.57±1.28	68.84±2.46	74.46±1.81	73.60±1.78
GLAD	91.86	89.08±1.13	88.45±1.54	89.65±1.16	89.26±1.34
RY	95.87	92.91±1.20	92.54±1.61	93.08±0.68	92.46±1.27
EM	92.28	89.87±1.16	88.32±1.84	90.45±0.96	90.11±0.85
PMLA	95.96	92.84±1.21	92.13±2.10	92.53±0.99	92.30±1.43
Ours	95.96	93.08±1.24	92.60±1.33	93.87±0.80	93.69±1.18
TRUE	100.0	95.79±1.28	92.76±2.38	96.23±0.81	96.30±0.96
Mem (MiB)	-	1,770	3,690	928	2,342
Time (s)	-	1,094.97	1,514.20	867.73	1,326.82

TABLE VI
PERFORMANCE ON SIMULATION DATASETS

Method	Road-G1		Road-G2		Road-G3		Road-G4	
	Label	Learning	Label	Learning	Label	Learning	Label	Learning
A ₁	95.32	89.08	80.20	76.96	86.47	82.74	93.83	88.29
A ₂	77.81	68.60	69.45	60.35	56.62	50.53	92.14	83.16
A ₃	61.89	56.58	56.62	46.95	51.14	41.84	92.74	85.72
A ₄	49.25	44.51	41.29	40.30	30.45	25.61	89.95	85.56
A ₅	25.57	21.16	30.65	29.63	17.91	24.38	90.25	84.61
GLAD	90.15	85.66	83.48	75.79	74.83	68.77	99.90	96.09
RY	94.63	86.49	87.76	79.25	80.90	71.43	100.0	96.10
EM	90.95	82.74	83.48	75.00	73.43	67.89	99.90	95.16
PMLA	96.42	90.22	88.96	78.42	86.27	82.32	100.0	95.86
Ours	96.42	90.72	88.96	81.26	86.27	82.95	100.0	96.84
TRUE	100.0	95.39	100.0	95.26	100.0	96.32	100.0	97.09

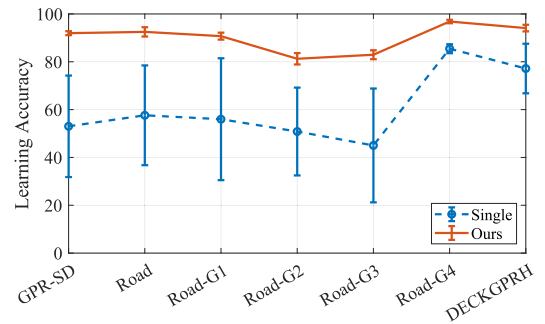


Fig. 10. Learning accuracy comparison, showcasing the superior performance of our multisource learning framework (Ours) over the single-label learning approach (Single).

aggregation stage. Meanwhile, using PMLA labels for deep image classification model training also achieved the highest accuracy. Introducing an LSR mechanism based on the

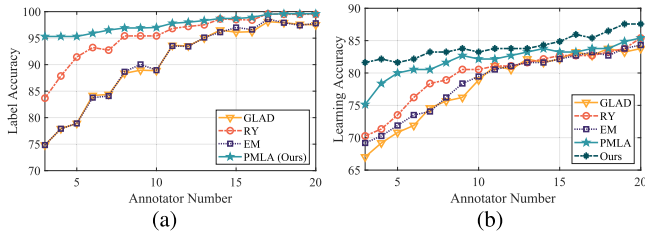


Fig. 11. Influences of different annotator numbers on the performance of the proposed method. (a) Label accuracy. (b) Learning accuracy.

probabilistic labels obtained by PMLA, the proposed method achieves even higher accuracy than learning with raw PMLA aggregated labels. In Road-G4, even when the accuracy of individual labels is already high, the use of multisource labels still significantly improves the training effectiveness of the model, especially when employing the proposed learning framework.

4) *Impact of Annotator Number*: In the practical task for GPR B-scans image recognition, it is common to engage a larger pool of annotators to train a more accurate and robust model. This article investigates the impact of the number of annotators on accuracy, comparing three common label aggregation mechanisms under the same conditions with varying annotator numbers. Added annotators' professional levels are randomly sampled from the original distribution.

Fig. 11(a) shows that the PMLA method's label accuracy significantly surpasses the other three methods when fewer annotators are used, exceeding 95% accuracy even with less than five annotators. As the number of annotators increases, other comparative methods also improve their label accuracy due to more knowledge sources but still underperform compared to PMLA. Comparing Fig. 11(a) and (b), it is clear that higher label accuracy results in better learning performance. Despite learning accuracy being lower than the corresponding aggregated label accuracy in Fig. 11(b), the trends are similar. The proposed method consistently outperforms the general PMLA-based learning model, regardless of the number of annotators. The benefits likely stem from the proposed method's full usage of annotation quality evaluation and analysis through imagewise reliability and LSR mechanisms.

5) *Comparison With Multisource Label Learning Baselines*: The proposed method is compared with several multisource label learning baselines including the following.

- 1) *DL-MV*: A CNN trained on the label fused by the label aggregation approach, as proposed in [48].
- 2) *CrowdLayer* [32]: A CNN augmented with an additional crowd layer that simultaneously incorporates labels from multiple annotators as supervision information, enabling the model to learn from the combined expertise of all annotators.
- 3) *TCL* [33]: A colabel learning method that learns with pseudo labels and alternates between optimizing the image classification model and the label aggregation model.
- 4) *TCL-MLP*: An enhanced version of TCL [33] that employs a neural network in place of the naive Bayes classifier to aggregate multisource labels.

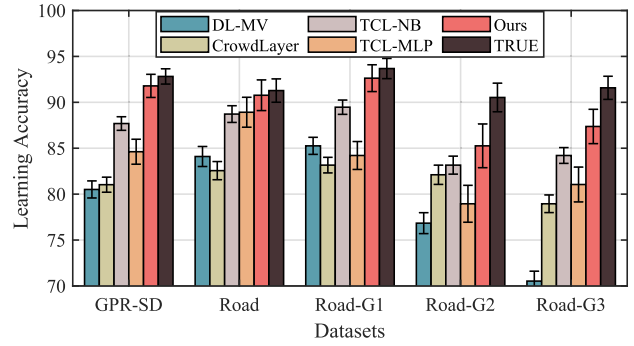


Fig. 12. Comparison with four DL-based multisource label learning baseline methods. The model that utilizes the ground truth for training (designated as TRUE) is presented as the upper bound since it employs the most accurate labels for learning rather than the erroneous ones.

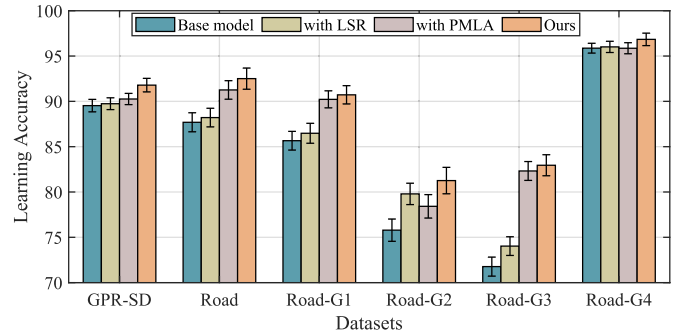


Fig. 13. Ablation study of the proposed method.

- 5) *TRUE*: A CNN training directly with the ground-truth labels, which implies the upper bound of the performance of the classification algorithm to some extent.

The comparison results are shown in Fig. 12.

It is observable that the proposed method substantially surpasses other baseline methods across all datasets. On (A) GPR-SD and (B) Road, two real-labeled datasets, and simulated dataset (D) Road-G1, the performance of our method closely approximates that of a model trained directly using real labels. On the other two datasets, there is still a gap between our method and the TRUE. This gap may be due to the fact that too many low-quality labels in these two datasets increase the difficulty of learning a good representation, which can also be verified from the same significant decline of other baselines.

6) *Ablation and Parameter Study*: The proposed model's strength is further investigated by examining each module's performance across five datasets, as shown in Fig. 13. The base model, lacking PMLA and LSR modules, uses direct normalization of multisource annotation to get a simple aggregation result. The results confirm that both the PMLA and LSR modules positively impact the model's final performance, with PMLA being especially beneficial.

Prior coefficient α for LSR and the imagewise reliability mapping temperature τ are key parameters. To study their impact, we set these parameters to various values and evaluated the learning accuracy, as shown in Fig. 14. Experiments reveal that the model's performance partly depends on suitable values for these two parameters. α regulates the LSR module's intensity. An overly large α may lead the model to focus on

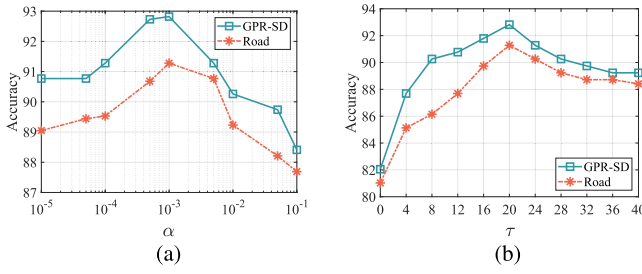


Fig. 14. Influence of the value of (a) α and (b) τ on the learning accuracy. The best value of α is 10^{-3} , and the best value of τ is 20. Both values are set as the default value.

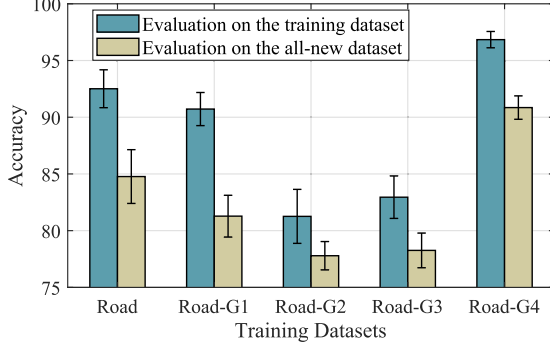


Fig. 15. Performances of the proposed method trained on the Road dataset and evaluated on an all-new dataset.

a specific label, disregarding errors in multisource labeling. The τ temperature hyperparameter of the Softmax function, which controls the probability gap between different labels, behaves similarly. Numerically, the model achieves the highest accuracy on both datasets when α is 10^{-3} , with τ set to 20. In addition, the learning accuracy consistently exceeds 87% and 80% on the respective datasets, which underscores the remarkable stability of the proposed method.

7) *Evaluation on a Completely New Dataset:* To provide valuable insights into the recognition accuracy and broader applicability of the proposed method, an additional experiment is designed to assess the performance of the trained model on a completely new dataset. The models are trained using images from the Road dataset and five sets of annotations and then evaluated on the all-new GPR-SD dataset without any further training. The results, as shown in Fig. 15, illustrate the model's ability to generalize across different datasets.

It is observed that due to the differences in data collection methods and geological environments of the collection locations, models trained on the Road dataset experience varying degrees of accuracy decline when using the all-new dataset. The trained model achieves an accuracy of 75% on the all-new dataset on all training datasets and reaches up to 90% when training on the Road-G4. This indicates that the proposed models still perform adequately on a completely new dataset.

8) *Discussions on the Physical Explanation:* As a geophysical imaging of Earth, the processing of GPR images obeys the fundamental physical principles. We evaluate the proposed method's capability to capture the physical insights with DNN during the learning process.

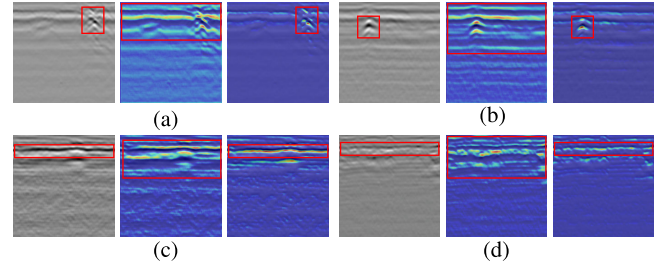


Fig. 16. Examples of GPR images with their Grad-CAMs of four object types. For each example, the left, middle, and right columns display the original GPR B-scan image, Grad-CAM produced by the traditional method, and Grad-CAM produced by the proposed method, respectively. (a) Example of cavity. (b) Example of pipeline. (c) Example of crack. (d) Example of loose.

Grad-CAM is a widely used technique to identify and highlight the key regions of an image that influences the neural network's decisions [49], [50]. To explore the physical basis of the proposed method in extracting key waveform features, four types of targets were selected from the Road dataset for comparison. The Grad-CAM visualizations from both the traditional single-label method and the proposed method were analyzed, with the results depicted in Fig. 16. The Grad-CAMs are derived from the first convolutional layer after 200 epochs of iteration. It is obvious that the proposed method more precisely captures the core physical waveform features of underground objects, which explains its superior performance. For example, in Fig. 16(b), the Grad-CAM of the proposed method primarily focuses on the hyperbola, which is the important feature that identifies pipelines [51]. In contrast, the Grad-CAM of the traditional method is distracted by irrelevant features, such as excessive noise, and displays scattered and cluttered key areas. This clutter significantly hampers the model's ability to extract and interpret electromagnetic radar signals effectively.

V. CONCLUSION

This article proposes a novel multisource label learning framework for GPR B-scan image classification in underground object recognition tasks to address the limitations of traditional DL methods. The framework uses multisource annotation benefits and alleviates annotation contradiction issues. It includes a PMLA mechanism to estimate label quality and identify potential correct labels from various annotations. An LSR module is also proposed to measure contradiction degrees and autoadjust regularization strength. Consequently, our method effectively leverages annotation information from specific subsurface areas by diverse-background annotators, leading to a more accurate classification model. Extensive experiments on annotated real-world GPR B-scan images reveal our method's positive effects in extracting potential correct labels and mitigating the risks associated with conflicting labels during learning.

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